University of Botswana

Department of Mathematics

Tutorial 2: Continuity of Functions

Part I: Continuity At a Point

In Exercises 1–4, discuss the continuity of the function at a point.

1. What three conditions must be satisfied for a function f to be continuous at a point c.

2. Determine whether or not the function is continuous at x=1.
$$f(x) = \begin{cases} x+1, & x<1 \\ 3, & x=1 \\ x^2+1, & x>1 \end{cases}$$

3. Determine whether or not the function is continuous at x=2
$$g(x) = \begin{cases} \frac{x^3-8}{x-2}, & x \neq 2 \\ 8, & x=2 \end{cases}$$

4. Show that the function is continuous at x=0.
$$f(x) = \begin{cases} 2x^2+4, & x \leq 0 \\ (x-2)^2, & x>0 \end{cases}$$

Part II: Continuity on a Closed Interval

In Exercises 5–7, discuss the continuity of the function on the closed interval.

5. What does it mean for a function to be continuous on a closed interval [a,b]?

6. $g(x) = \sqrt{49-x^2}$ on $[0,5]$

7. $g(x) = \begin{cases} 3-x, & x \leq 0 \\ 3+\frac{1}{2}x, & x>0 \end{cases}$ on $[-1,4]$

Part III: Discontinuities

In Exercises 8–13, find the x-values (if any) at which f is not continuous. Which of the discontinuities are removable?

8. $f(x) = \frac{6}{x}$
9. $f(x) = x^2 - 4x + 4$
10. $f(x) = \frac{x}{x^2+1}$
11. $f(x) = \frac{x+2}{x^2-x-6}$
12. $f(x) = \frac{|x-5|}{x-5}$
13. $f(x) = \begin{cases} x, & x \leq 1 \\ x^2, & x > 1 \end{cases}$

Part IV: Making a Function Continuous

In Exercises 14–17, find the constant a, or the constants a and b, such that the function is continuous on the entire real number line.

14. $f(x) = \begin{cases} 3x^2, & x \geq 1 \\ ax - 4, & x < 1 \end{cases}$
15. $g(x) = \begin{cases} 4 + \sin 2x, & x < \pi \\ \\ a - 2\cos x, & x \geq \pi \end{cases}$
16. $g(x) = \begin{cases} \frac{x^2-a^2}{x-a}, & x \neq a \\ 8, & x = a \end{cases}$
17. $f(x) = \begin{cases} 2, & x \leq -1 \\ ax + b, & -1 < x < 3 \\ -2, & x \geq 3 \end{cases}$

Part V: Intermediate Value Theorem Applications

In Exercises 18–20, apply the Intermediate Value Theorem as indicated.

18. State the Intermediate Value Theorem.
19. Explain why $f(x) = x^3 + 5x - 3$ has a zero in $[0, 1]$.
20. Use Intermediate Value Theorem to approximate a zero of $f(x) = x^3 + x - 1$ in $[0, 1]$.